

## Case 41: Custom fuel-parameter Rothermel sweep

### 1 Verification purpose

This case verifies ELMFIRE’s public custom fuel-table interface and the Rothermel response to one-at-a-time changes in dead 1-h load, 1-h SAV, fuel-bed depth, dead extinction moisture, and heat content. It can expose column misalignment, ignored custom values, stale precomputed coefficients, or unit conversion errors. The question is whether every generated code-250 row produces independently calculated reaction intensity and ROS within 1% and 2%.

### 2 Mathematical formulation and numerical implementation

The custom row supplies class loads  $w_0$ , surface-area-to-volume ratios  $\sigma$ , depth  $\delta$ , dead extinction  $M_x$ , and heat content  $h$ . These feed packing ratio  $\beta = \sum w_0/(\delta\rho_p)$ , optimum packing, reaction velocity, propagating flux, damping, heat sink, and

$$R_0 = I_R\xi/(\rho_b\epsilon Q_{ig}). \quad (1)$$

The formulation follows Andrews (2018), RMRS-GTR-371. Particle density and mineral constants are fixed by the inspected ELMFIRE interface and therefore are not claimed as user-sweepable variables.

### 3 Derivation of expected behavior

The baseline row is copied and exactly one public column is changed in each variant. A case-local fuel table is written beside that variant and read independently by the oracle. The full equation is recalculated rather than imposing a monotonic trend: load, SAV, and depth have competing effects through packing and reaction velocity, so a simple ordering would be scientifically unjustified.

Each custom-fuel reference is recalculated from  $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$ , changing only the indicated fuel-table column. The substitutions produced from the generated table are

| variant         | $R_{\text{ref}}$ (m min <sup>-1</sup> ) | $I_{R,\text{ref}}$ (kW m <sup>-2</sup> ) |
|-----------------|-----------------------------------------|------------------------------------------|
| baseline        | 0.411242                                | 1045.499                                 |
| $w_{1h} = 0.05$ | 0.253350                                | 688.121                                  |
| $w_{1h} = 0.15$ | 0.551674                                | 1379.611                                 |
| $SAV = 1000$    | 0.241692                                | 960.426                                  |
| $SAV = 3000$    | 0.662626                                | 1074.561                                 |
| $d = 0.5$       | 0.225975                                | 910.468                                  |
| $d = 2.0$       | 0.694647                                | 991.942                                  |
| $M_x = 15\%$    | 0.371790                                | 945.198                                  |
| $M_x = 35\%$    | 0.438500                                | 1114.797                                 |
| $h = 6000$      | 0.308432                                | 784.125                                  |
| $h = 10000$     | 0.514053                                | 1306.874                                 |

For example, replacing the baseline 1-h load 0.10 by 0.05 lb ft<sup>-2</sup> changes  $R_0 = 1.349221$  to 0.831202 ft min<sup>-1</sup>; multiplication by 0.3048 gives 0.253350 m min<sup>-1</sup>. Each variant uses  $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$  and the analogous  $e_I$ . The TOA fit  $t = a + bx$  gives

$R_{\text{TOA}} = 60/b$  and  $R^2 = 1 - SSE/SST$ . The aggregate substitutions are  $\max e_R \leq 0.02$ ,  $\max e_I \leq 0.01$ ,  $\max e_{\text{TOA}} \leq 0.05$ , and  $\min R^2 \geq 0.995$ .

## 4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Variants predicted at or above  $0.10 \text{ m min}^{-1}$  additionally require TOA-derived ROS error at most 0.05 and minimum  $R^2 = 0.995$ ; slower variants are assessed from direct ROS and intensity. Pointwise equality to a complete recalculation detects both ignored columns and wrong nonlinear propagation. The baseline and ten low/high variants are all required.

## 5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported  $[-1, 1]$  interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, SIMULATION\_DT and SIMULATION\_DTMAX use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the SIMULATION\_TSTOP in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. All variants use fuel code 250, 5/7/9% dead moisture, 90% live moisture, zero wind and slope, and 10 m cells. A finite ignition strip supplies a planar east face in the interior analysis corridor while remaining clear of the lateral grid boundaries. The baseline row uses 0.10/0.05/0.02 dead loads, 0.08 live-woody load, 1-h SAV  $2000 \text{ ft}^{-1}$ , depth 1 ft, dead extinction 25%, and heat content  $8000 \text{ Btu lb}^{-1}$ . Each output should match its own oracle; no universal increasing trend is predicted.

| Parameter        | Configured value                                    | Units              | Role                             |
|------------------|-----------------------------------------------------|--------------------|----------------------------------|
| Domain/grid      | 1400 m square; 10                                   | m/cell             | Planar interior propagation      |
| Custom fuel code | 250                                                 | code               | Selects variant-local table row  |
| Changes          | load, SAV, depth, extinction, heat                  | native table units | One-at-a-time interface exercise |
| Wind/slope       | 0 / 0                                               | mph / degrees      | Isolates custom fuel response    |
| Timing           | generated whole-second step and exact-multiple stop | s                  | Terminal-evidence compatibility  |

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells

selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/load\_high; WS=0 mph, WD=270 deg

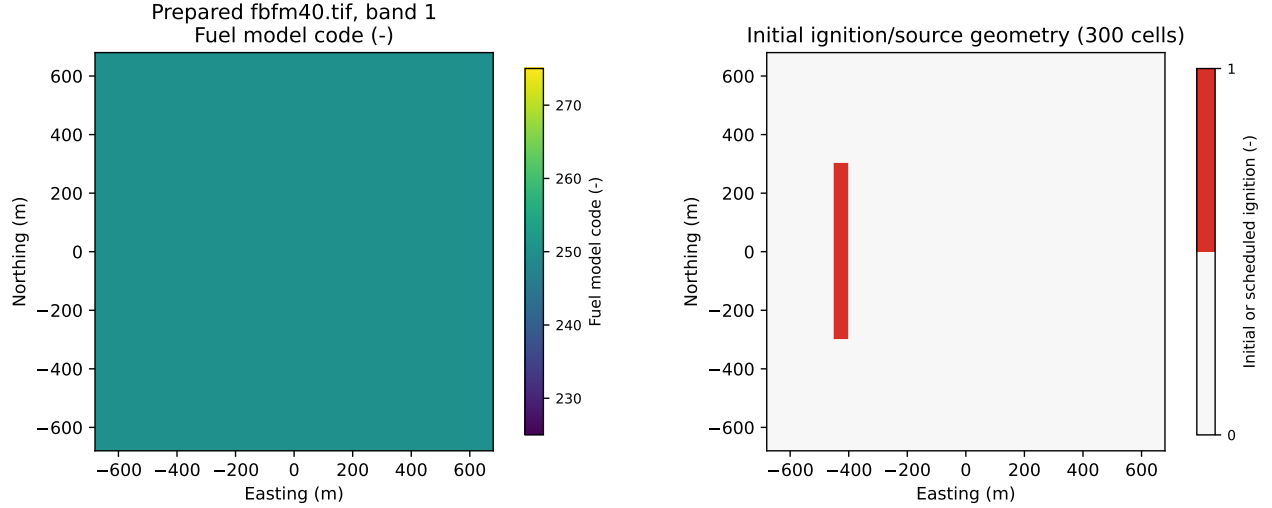


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from  $\phi \leq 0$ .

## 6 Actual simulation results

The generated record contains 11 of 11 variants and reports **PASS**.

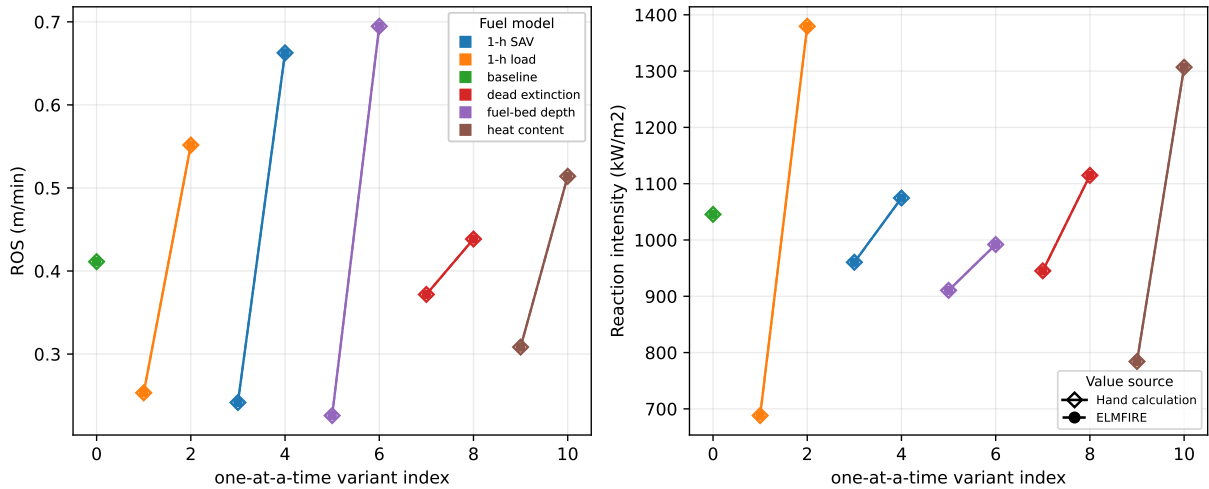


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the custom fuel-table variants. Each comparison pair uses the same color.

## 7 Calculated metrics and verification decision

| Metric                                    | Limit        | Calculated             | Status |
|-------------------------------------------|--------------|------------------------|--------|
| maximum direct-ROS relative error         | $\leq 0.02$  | 4.62672e-07 fraction   | PASS   |
| maximum reaction-intensity relative error | $\leq 0.01$  | 1.10118e-07 fraction   | PASS   |
| maximum TOA-ROS relative error            | $\leq 0.05$  | 0.0151881 fraction     | PASS   |
| minimum TOA regression R2                 | $\geq 0.995$ | 0.999968 dimensionless | PASS   |

The authoritative decision is **PASS**.

## 8 Technical interpretation

If only one low/high pair fails, its fuel-table column or dependent coefficient is suspect. A common intensity error suggests table parsing or units; correct intensity with wrong ROS points to heat sink, density, or flux ratio. Identical results across a changed pair indicate the column was ignored.

## 9 Scope and limitations

This case tests only parameters exposed by the current custom table and one synthetic baseline. It does not test particle density, mineral contents, spatial mixtures, dynamic curing, or physical realism of the synthetic fuel.